

Wenlong Ding

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EDUCATION

The Chinese University of Hong Kong (CUHK)

Shatin, Hong Kong, China

Ph.D. in Computer Science and Engineering

09/2021 – Present

- GPA: very good | Rank: very good

Huazhong University of Science and Technology (HUST)

Wuhan, Hubei, China

B.E. in Computer Science and Technology

09/2017 – 06/2021

- GPA: 90.57/100 or 3.96/4.0 | Rank: Top 5%
- English Level: CET6-612

RESEARCH EXPERIENCE & ACADEMIC COMPETITION

Research Intern, Retention-aware Container problem in edge computing

04/2019 – 06/2019

Research Intern in HUST, Advised by Prof. Fangming Liu

HUST, China

- We provide a model to handle C2RD (container caching jointly with request distribution) question, which finds the tradeoff between service latency cost and container retention cost.
- Inspired by the classic ski-rental problem, we present RDC (Random Distribution and Caching) algorithm for multi-node case at one timeslot (a special case).
- We propose O-RDC (Opportunistic-RDC) to handle the general cases (multi-node case at consequent timeslots). It turns out that the model has a relatively good competitive ratio.
- We carry out an experiment on a real-world dataset (EUA dataset). The model outperforms existing cache strategies by up to 97% in overall system cost.

The Opioid Crisis, Problem C of Mathematical Contest in Modeling 2019

01/2019

This project won the Meritorious Winner awarded by COMAP in 04/2019

COMAP, USA

- We should figure out the spreading source and future users distribution of opioid with the given dataset.
- Use SPBFA (Source-Finding Problem Based on Floyd Algorithm) to find the spreading source.
- Determine the edges weights using PCCM (Partial correlation coefficient modified function).
- Use Granger Causality test to extract more possible factors which will influence our model. Combined with PCCM, we recompute the weights in our previous model.
- Use Markov Chain to predict the future trend of opioid.

PROJECTS

Disk Failure Prediction Via GAN Anomaly Prediction

09/2020 – 10/2020

- Convert one-dimensional SMART data into 2D SMART data through sliding windows.
- In the training phase, only the 2D SMART data of the normal disk is trained in GAN anomaly prediction module to capture its data characteristics, and the threshold is set according to its Loss.
- In the verification phase, normal disks and abnormal disks are distinguished through the set threshold.
- Achieve a FDR as high as 95% and a FAR as low as 1% in a real world dataset *Backblaze*.

Mini-C Language compiler with Performance Optimization

05/2020 – 06/2020

- The system translates Mini-C Language (Subset of C Language) to MIPS Assembly language.
- The system can identify all lexical and grammatical errors through lexical and syntax analysis program. Also, the static semantic program can identify 20+ kinds of static semantic errors.
- Construct DAG to achieve local optimization of target code and extract invariant operations to achieve loop optimization. The length of target code has been reduced by 31.3% in the benchmark.

- Implement target code register allocation strategy using DAG heuristic sorting algorithm. The number of registers we use has been reduced by 22.8% in the benchmark.

Performance Optimized Five-stage Pipeline MIPS CPU

03/2020

- Construct controller, pipeline interface components, data path, etc. to form a Pipeline CPU.
- Add data redirection technology and dynamic branch prediction mechanism (BHT prediction module), the runtime of benchmark has been reduced by 36.6% and 49.8% respectively.
- Support two modes of interrupt in our pipeline: hardware interrupt, software interrupt.
- Implement a box pushing game with graphic interface and run on the CPU completed above.

A Web Server Software Supporting Multithreading

12/2019

- Support to configure the listening address, listening port and home directory of the web server.
- Web Server can return the requested page normally, and handle exceptions to error pages.
- Support the transmission of multiple types of files, such as multimedia files.
- Support multi-thread communication and GUI.

OVERSEAS EXCHANGE EXPERIENCE

Participant of Summer Workshop

07/2019

School of Computing (SoC), National University of Singapore (NUS), Singapore

- Conduct a data processing project about predicting the house price with the dataset in Kaggle.
- Our model gives a good result with RMSE as low as 0.114 and ranked top 4% in Kaggle project.

Academic Course About Deep Learning

10/2018

University of California, Los Angeles(UCLA), USA

- Take some courses about the basic knowledge about Deep Learning.
- Conduct a project using CNN to identify the face dataset and the accuracy can reach 98%.

HONORS AND AWARDS

09/2020	Merit Student, HUST
12/2020	Outstanding Individual in University Students Innovation and Entrepreneurship Project, HUST
09/2019	Scholarship for Extracurricular Activities and Sports, HUST
04/2019	Meritorious Winner, Mathematical Contest in Modeling, COMAP, USA
03/2018	Freshmen Scholarship, HUST

TECHNICAL SKILLS

Languages: C/C++, Python, html/css/javascript

Software: VScode, Xcode, Microsoft Office, Vivado, Logisim, QT Creator, SQL Server management

Operating System: MacOS, Linux(ubuntu), Windows10